



AMARAPINNI VENKATA SAI VIGNESH

av2684@srmist.edu.in | 7396581364 | 2027

B.Tech, Computer Science and Engineering

Address : Tiruchirappalli Campus srm nagar, trichy,Chennai-highway,Tamil Nadu 621105



Education

SRM IST-Trichy

B.Tech · Computer Science and Engineering

2027

CGPA - 8.74/10

Dr junior college

Class XII - State Board · MPC · Ayyappa Swamy Temple, Thagarapuvalasa road, Pendurthi, Rajayyapeta, Andhra Pradesh 531173

2023

Percentage - 56%

CHAITANYA HIGH SCHOOL

Class X - State board · Ukkunagaram Vsp, Visakhapatnam district

2021

Percentage - 94%

Experience

Defence Research & Development Organisation (DRDO) – CVRDE

Project Intern · Intern · Internship

Dec 2024 - Present

Avadi, Chennai, Tamil Nadu

Completed an internship at DRDO – Combat Vehicles Research & Development Establishment (CVRDE), Avadi, under the guidance of Scientist 'E'. Worked on the project "Development of Computer Aided Process Planning (CAPP) Module for Machining of Prismatic Components Using Feature-Based Recognition."

Developed a customized Python program to extract data from AutoCAD DXF files and convert it into text format. Implemented unit testing, automated testing, and quality assurance checks. Enabled automatic generation of G-code and M-code for precise machining operations.

Projects

Development of a CAPP Module for Machining of Prismatic Components |

Dec 2024 - Present

CERTIFYD BY DRDO,CVDE

SRM Institute of Science and Technology ,Internship at DRDO CVRDE, Avadi ·
Python; AutoCAD DXF; Pandas; NumPy; pytest (unit testing); Git; G-code/M-code
generation scripts

Machine Learning; CAD/CAM; Process
Planning

Project overview: Developed a Computer-Aided Process Planning (CAPP) module to automate process planning for prismatic components using feature-based recognition. The system reads geometry from AutoCAD DXF files, converts features to structured text, and automatically generates validated G-code and M-code for CNC machining.

Smart Event Management System

Dec 2025 - Jan 2026

SRM Institute of Science and Technology · React.js, TypeScript, Vite, Tailwind CSS,
Supabase, PostgreSQL, Supabase Auth, React Hook Form, Zod, jsPDF, Recharts,
Vitest, ESLint

Full Stack Web Development

Project overview: I Have Developed a Full Stack Smart Event Management System to simplify college event management and student participation. Implemented secure role-based authentication using Supabase Auth, enabling administrators to manage events and approve registrations. Built features including event registration, participation tracking, room allocation, color-coded room availability, automated OD letter generation, and PDF report export. Developed a responsive user interface using React.js, TypeScript, Vite, and Tailwind CSS, with Supabase and PostgreSQL powering the backend, authentication, and database operations. Integrated Recharts to provide interactive dashboards and event analytics.

Certifications

Android Developer Virtual Internship | [Link](#)

GOOGLE · a677ee6378b24b8859c63b54c82f7336

During my Android Developer Virtual Internship, I learned the fundamentals of Android app development using **Kotlin/Java** and Android Studio.
I designed user interfaces using XML and developed basic applications such as a login system, calculator, and to-do list app. I implemented user interactions, navigation between activities, and handled form validations.
I also worked with REST APIs to fetch and display data in applications and learned database integration using SQLite and Firebase.
Additionally, I understood core concepts like MVVM architecture, debugging, and version control using Git.

Sept 2025

AI-ML Virtual Internship | [Link](#)

GOOGLE · 4f2d18d3243af0654d3b0539d7191cb3

Completed an AI-ML Virtual Internship gaining practical experience in Python and TensorFlow. Worked on data preprocessing, machine learning model development, and basic deployment concepts while understanding supervised and unsupervised learning techniques.

Jun 2025

Volunteering

SRM Institute of Science and Technology, Tiruchirappalli | [Link](#)

Volunteer · Innovation & Entrepreneurship Development

I have been a volunteer at the Innovation, Design and Entrepreneurship (IDE) Bootcamp – Edition 2, Phase I, organized by AICTE and the Ministry of Education's Innovation Cell at SRM IST, Tiruchirappalli. Contributed to event coordination, participant support, and smooth execution of innovation and entrepreneurship sessions. Enhanced leadership, teamwork, communication, and event management skills while gaining exposure to startup and innovation ecosystems.

Sept 2024 - Sept 2024

Competitions

PSG Institute of Technology and Applied Research | [Link](#)

participated · PSG Institute of Technology and Applied Research

Attended REAL ASCEND, a workshop held during the national technical symposium YUKTA'24 at PSG Institute of Technology and Applied Research. Participated in hands on sessions and technical discussions focused on practical skills and emerging tools relevant to engineering students. Recognized for active participation and awarded a certificate

Mar 2024

Conferences and Workshops

REAL ASCEND — Workshop | [Link](#)

PSG Institute of Technology and Applied Research · Neelambur, Coimbatore 641062

Attended **REAL ASCEND**, a workshop held during the national technical symposium **YUKTA'24** at PSG Institute of Technology and Applied Research. Participated in hands on sessions and technical discussions focused on practical skills and emerging tools relevant to engineering students. Recognized for active participation and awarded a certificate

Mar 2024

Extra Curricular Activities

NCC A,B,C(CURRENT) | [NCC](#)

Actively participated in National Cadet Corps (NCC) activities, developing leadership, discipline, and teamwork skills. Engaged in drills, camps, and social service initiatives that enhanced physical fitness, responsibility, and organizational abilities.

Sept 2023 - Feb 2026

Skills

Programming Languages: full-stack, Python,JavaScript, C++, HTML, CSS,SQL. Frontend: Figma. Backend: Python,G-code, M-code,. Tools: Git, GitHub, VS Code,AutoCAD .

Languages

ENGLISH,HINDI,TELUGU,TAMIL [Professional Working Proficiency]

Links

[LinkedIn](#), [GitHub](#)